Assignment 6: Public Key Cryptography DESIGN.pdf:

Description of Program:

This assignment will require three programs - a key generator(keygen), an encryptor(encrypt), and a decryptor(decrypt). The keygen program will be in charge of key generation, producing RSA public and private key pairs. The encrypt program will encrypt files using a public key, and the decrypt program will decrypt the encrypted files using the corresponding private key.

Files to be included in directory “asgn6” + pseudocode/structure:

- decrypt.c: This contains the implementation and main() function for the decrypt program.
- encrypt.c: This contains the implementation and main() function for the encrypt program.
- keygen.c: This contains the implementation and main() function for the keygen program.
- numtheory.c: This contains the implementations of the number theory functions.
- numtheory.h: This specifies the interface for the number theory functions.
- randstate.c: This contains the implementation of the random state interface for the RSA library
- and number theory functions.

- randstate.c: This specifies the interface for initializing and clearing the random state.
- rsa.c: This contains the implementation of the RSA library.
- rsa.h: This specifies the interface for the RSA library.
- Makefile
    - CC = clang must be specified.
    - • CFLAGS = -Wall -Wextra -Werror -Wpedantic must be specified.
    - • pkg-config to locate compilation and include flags for the GMP library must be used.
    - • make must build the encrypt, decrypt, and keygen executables, as should make all.
    - • make decrypt should build only the decrypt program.
    - • make encrypt should build only the encrypt program.
    - • make keygen should build only the keygen program.
    - • make clean must remove all files that are compiler generated.
    - • make format should format all your source code, including the header files.
- README.md
    - In proper Markdown syntax.
    - Describe how to use your program and Makefile
    - Lists and explains any command-line options that program accepts.
    - Reports any false positives by scan-build

    - Notes down any known bugs or errors in this file as well for the graders.
- DESIGN.pdf
    - This document must be a proper PDF.
    - Describes design and design process for program with enough detail such that a sufficiently knowledgeable programmer would be able to replicate implementation.